

Benjamin (Benji) Adjepong

U.S. CITIZEN, ELIGIBLE TO OBTAIN CLEARANCE

(757)-319-9675

benji@vt.edu

linkedin.com/in/benjaminadjepong

Chesapeake, VA

EXECUTIVE SUMMARY

Goal-driven Computer Engineering student with a keen interest in embedded systems & robotics. Proficient in programming languages such as C/C++ and Embedded C, and experienced in microcontroller programming. Seeking experiential learning opportunities such as internships/co-ops to learn and apply technical skills in a collaborative environment (Summer 2024)

EDUCATION

B.S. COMPUTER ENGINEERING - VIRGINIA TECH | BLACKSBURG, VA

GPA: 3.55

EXPECTED SPRING 2025

Major: Machine Learning

Secondary Focus: Controls, Robotics & Autonomy

Minor: Computer Science

RELEVANT COURSEWORK: Data Structures & Algorithms (C++), Embedded Systems, Computer Architecture & Organization, Continuous & Discrete Systems, Signals & Systems, Linear Algebra, Discrete Mathematics, Physical Electronics

WORK EXPERIENCE

COLLINS AEROSPACE | STERLING, VA

JUNE 2023 – AUGUST 2023

Mission Systems Software Engineer Intern – Station Keeping Playbook

- Developed a low-cost operational testing platform on ROSMASTER X3 ground robots functioning as an alternative to costly flight testing for unmanned aerial systems, ultimately reducing traditional testing costs by approximately 90%
- Integrated Collins Aerospace autonomous flight software onto ROSMASTER ground robots using Python-C bindings to reliably test the Station Keeping Playbook autonomous capabilities for unmanned aerial systems
- Utilized various python libraries to process GPS & IMU data and command motors based on Station Keeping outputs

HUME CENTER FOR NATIONAL SECURITY & TECHNOLOGY | BLACKSBURG, VA

DECEMBER 2022 – MAY 2023

Paid Undergraduate Research Assistant – Boeing IoT Device Characterization

- Conducted extensive experiments testing range, battery and durability of Bluetooth Low Energy (BLE) IoT devices
- Collected and parsed through raw data using Putty and Python scripting
- Compiled detailed reports on the findings of the experiments using Microsoft Word and Excel

PROJECTS

AUTONOMOUS TRACTOR | ARDUINO, PYTHON

SPRING 2023

- Developed graphical user interface (GUI) for an autonomous tractor using Python and PyGame
- GUI utilized Bleak Client libraries to send a receive data from the tractor via Bluetooth Low Energy (BLE)
- Interfaced Arduino Uno with 4+ sensors, enabling the tractor to traverse a course autonomously

NANOGAMES | EMBEDDED C

SPRING 2023

- Developed series of minigames on TI MSP432 microcontroller board
- Minigames utilized several hardware abstraction layers to configure input and output for joystick, buttons and LEDs

DRONE FLIGHT PLAN | C++

FALL 2022

- Implemented finite state machine with 5+ states to parse text commands successful drone operation
- Executed flight plans with drone avoiding obstacles, performing 3+ different maneuvers

SKILLS

LANGUAGES

- Embedded C, C, C++, Python, MATLAB, Verilog/VHDL

LIBRARIES

- PyGame, ctypes (Python-C Bindings)

PROGRAMS

- Visual Studio Code, TI Code Composer Studio, LTspice, SolidWorks, CCS, Putty, Figma, Microsoft Word/Excel/PowerPoint

TECHNICAL TOOLS

- Arduino, Nvidia Jetson Nano, ROSMASTER X3, TI MSP432 Launchpad & Boosterpack, AD2 Oscilloscope, DE10-Lite FPGA

EXTRACURRICULAR

VIRGINIA TECH COMPETITIVE ROBOT ORGANIZATION (VT CRO)

2023 – PRESENT

- Developed an autonomous robot to compete in the IEEE regional competition

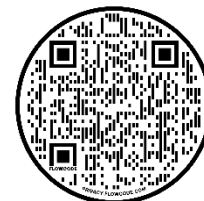
NATIONAL SOCIETY OF BLACK ENGINEERS (NSBE)

2021 – PRESENT

- Participated in 10+ professional development events, e.g., professional etiquette workshops

INTERESTS & HOBBIES

- **INTERESTS:** Embedded Software Systems, Robotics, Machine Learning
- **HOBBIES:** E-skating, Fitness, Ping-Pong, Basketball, Soccer



LinkedIn